

A Measure-Theoretic Discrete Curvature Framework for Metabolic Gene Sensitivity: From Active-Set Geometry to Transcriptional Response

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Abstract

Parametric flux balance analysis defines, for every metabolic network, a piecewise-affine map from environmental and genetic parameters to optimal fluxes. We show that the distributional second derivative of this map is a matrix-valued Radon measure concentrated on the codimension-one strata of the active-set complex, and that this measure — not a smooth curvature — is the canonical carrier of the network’s rerouting geometry: loop holonomy scales linearly in loop size (slope 1.00, versus the quadratic scaling of any smooth map), and the second-order response mass concentrates (100.0000%) on operational active-set switches. We prove a refinement–resolution bridge (Theorem B’): under mesh refinement the atomic measure converges weakly to the smooth curvature density, the dual-cell reconstruction converges strongly in L^1 , and the total-variation claims of earlier formulations are replaced by an exact counterexample calculus; a resolution parameter σ separates the discrete and smooth regimes through the measured window $h \ll \sigma \ll L_{\text{var}}$. From the measure we derive a gene-level sensitivity metric κ^μ (measure mass along a parametric trajectory) and show, on *E. coli* (M3D microarrays, $n = 424$ genes), that κ^μ predicts carbon-depletion transcriptional response ($r = +0.395$, $p = 2.6 \times 10^{-17}$; partial $r = +0.269$ controlling reference level; metric-invariant $\rho = 0.99998$ against the lexicographic definition), while the association does not propagate to the protein layer — consistent with a model in which cells transcribe the potential for rerouting while buffering its translation. Double-knockout epistasis aligns with active-set structure ($\rho_S = 0.865$), and genotype loops close with phenotypic memory (66% non-reverting). The framework unifies three previously distinct sensitivity objects as one measure at different resolutions.

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1 Introduction

The object. Let a metabolic network with m reactions be operated by flux balance analysis with parameters $\theta \in \Theta \subset \mathbb{R}^p$ (uptake bounds, knockdown scalings) entering the constraints affinely. The optimal flux map $v : \Theta \rightarrow \mathbb{R}^m$ is continuous and piecewise affine (parametric linear programming), with breakpoints on the active-set strata of the constraint complex. Its distributional second derivative $\mu = D^2v$ is therefore not a function but a *measure* — the discrete curvature carrier of the network’s rerouting geometry.

The claim structure. This manuscript establishes, in order:

- (i) **[R1]** the atom structure and weak convergence of μ_h under refinement (Theorem B', §4), with the exact counterexample calculus that separates what is provable from what is false in earlier formulations;
- (ii) **[R2]** the two-layer separation: the codimension-one Hessian layer is the unbiased curvature estimator; the codimension-two angle-defect layer converges to the Gauss-map area density with an exact sec-law bias (Proposition 4.3);
- (iii) **[R3]** the regime dichotomy: piecewise-affine maps with fixed kink complexes produce $O(\varepsilon)$ loop holonomy (measured slope 1.00) where smooth maps produce $O(\varepsilon^2)$;
- (iv) **[R4]** the value function $\Phi = c_{\text{bio}}^\top v^*$ is the canonical tie-break-free carrier, with shadow-price jumps measurable to 6–7 digits against Danskin envelopes;
- (v) **[E-series]** the measure-derived metric κ^μ predicts carbon-depletion transcriptional response and is buffered at the protein layer (§6).

Positioning. Kochanowski et al. demonstrated that global metabolic coordination under catabolic limitation is largely passive at the protein/metabolite level, driven by Crp and local metabolite concentrations. Our work asks a different question: whether a gene-specific, network-derived rerouting necessity predicts transcriptional response. We find that it does, and that this transcript-level association does not propagate to the protein layer — consistent with a model in which cells transcribe the potential for rerouting while buffering its translation.

2 Notation protocol and the R/D/E hierarchy

- **[R- n]** blocks are proved results (theorems, propositions, lemmas) or machine-verified claims with reproducible artifacts.
- **[D- n]** blocks are definitions, constructions, and design decisions (metric definitions, engine specifications).
- **[E- n]** blocks are executed experiments with deposited artifacts (data files, figures, statistics).
- Conjectures are labelled as such; plausible-but-unproven claims are demoted, never promoted.

3 The discrete curvature measure of parametric FBA

3.1 [D1] The measure

Definition 3.1 (Atomic curvature measure). Let $u : \Omega \rightarrow \mathbb{R}^m$ be continuous and piecewise affine on a polyhedral complex $\mathcal{C}$ covering Ω . For each codimension-one facet F of $\mathcal{C}$ shared by cells τ^- ,

τ^+ , let $[\nabla u]_F = \nabla u|_{\tau^+} - \nabla u|_{\tau^-}$, let n_F be the unit conormal oriented $\tau^- \rightarrow \tau^+$, and let $|F|$ be the facet measure. The *atomic curvature measure* is the matrix-valued Radon measure

$$\mu = D^2 u = \sum_F A_F \delta_F, \quad A_F = [\nabla u]_F \otimes n_F |F| \in \mathbb{R}^{m \times p \times p}.$$

Remark 3.2 (Support). μ is concentrated on the $(p-1)$ -skeleton. The $(p-2)$ -skeleton carries a different measure (the angle-defect layer, Proposition 4.3); the two must not be conflated. In $p=1$ the atoms are the slope jumps at breakpoints.

3.2 [D2] The measure-theoretic sensitivity metric

Definition 3.3 (κ^μ). For a C^1 parameter trajectory $\theta(t)$, $t \in [0, T]$, and its piecewise-affine optimal-flux response $v(\theta(t))$, let $\{t_e\}_e$ be the event times at which the trajectory crosses active-set strata, with atoms $A_e = A_{F(e)}$ evaluated on the crossed facet. Define

$$\kappa^\mu(g) = \frac{1}{T} \sum_e \|A_e\| w_e(g),$$

where $\|\cdot\|$ is the entrywise 1-norm on $\mathbb{R}^{m \times p \times p}$ and $w_e(g) \in \{0, 1\}$ selects the events whose crossed stratum changes the activity of reactions assigned to gene g (max over its reactions, per the GPR association).

Remark 3.4 (κ^μ is the safe regime of the reconstruction). κ^μ integrates measure mass along a one-dimensional trajectory — the telescoping regime in which total variation and mass are automatically consistent (R2 remarks in §4); its resolution independence (measured mass 288.77 stable under $4 \times 8 \times$ trajectory refinement) is exactly the L^1 -reconstruction regime of Theorem B'(iv).

3.3 [D3] The value-function layer (Alexandrov)

Proposition 3.5 (Existence and tie-break-freeness). *Let $\Phi(\theta) = \max\{c^\top v : Sv = 0, \ell(\theta) \leq v \leq u(\theta)\}$ with ℓ, u affine, feasible and bounded. Then Φ is concave (pointwise infimum of affine dual objectives over a θ -independent dual feasible set), and $-\Phi$ carries a symmetric matrix-valued Radon Hessian measure $D^2(-\Phi)$ (Alexandrov), independent of the optimal-flux selection rule. Concavity fails in general under OR (isoenzyme) GPR rules for simultaneous multi-gene capacity vectors; single-gene axes, exchange-bound families, and AND-only subnetworks retain it.*

Proposition 3.6 (Two-layer structure of $D^2(-\Phi)$). *Let f be convex piecewise affine. (i) The singular part of $D^2 f$ is supported on the codimension-one facets, with density $[\partial_n f](g^+ - g^-) \otimes n$ per unit facet measure; it has no atoms: the mass in an ε -neighborhood of a codimension-two vertex is $O(\varepsilon)$ (measured log-log slope 1.000). (ii) The Monge–Ampère measure $\det D^2 f$ is atomic at codimension-two vertices, each atom equal to the volume of the normal fan $|\text{conv}\{\nabla f(\tau) : \tau \ni \text{vertex}\}|$ (measured: constant 0.5 under ε -shrinking).*

Proposition 3.7 (Dual optimal face identity). *At a codimension-two vertex θ_v of a parametric-LP value function, the Monge–Ampère atom of $-\Phi$ equals the area of the normal fan, and the fan is the image of the LP dual optimal face under $\theta \mapsto U^\top y$:*

$$\det D^2(-\Phi)(\{\theta_v\}) = |\text{conv}\{\nabla C_i\}| = |\{U^\top y : (\lambda, y) \text{ optimal dual}\}|.$$

Machine-verified on a two-parameter max-flow LP (the phenotype phase plane in miniature): vertex refined to machine precision by exact triple-tie solve, $\Phi(\theta_v)$ matching the LP value to 10^{-15} , fan area 0.235 vs dual-face enumeration area 0.235000068 (16 direction LPs over the dual optimal face).

Remark 3.8 (Layer separation and PhPP positioning). The phenotype phase planes of Edwards–Ibarra–Palsson are the chamber complex of Φ ; their sector corners are the atoms of the Monge–Ampère layer. The empirical κ^μ of Definition 3.3 is a functional of the *flux-map* measure D^2v (Definition 3.1), which is tie-break-sensitive: in a controlled follower-variable LP, the value function stays affine and identical under two lexicographic tie-breaks while the follower’s second-difference mass flips $0.0100 \leftrightarrow 10^{-13}$. The controlled relation between the value layer and the flux layer is stated as Conjecture 4.9; the association results of §6 stand on the flux layer alone.

4 The refinement–resolution bridge (Theorem B')

Throughout this section $\Omega \subset \mathbb{R}^p$ is compact, $p \geq 1$, $u \in C^2(\Omega; \mathbb{R}^m)$, $\{\mathcal{T}_h\}$ is a shape-regular family of conforming triangulations with mesh size $h = \max_\tau \text{diam}(\tau) \rightarrow 0$, and u_h is the continuous piecewise-affine interpolant of u on $\mathcal{T}_h$.

Theorem 4.1 (Refinement–resolution bridge, corrected form). *Let $\mu_h = D^2u_h$ (Definition 3.1) and $\mu = D^2u$ dvol. Then:*

- (i) **(Atoms / support.)** μ_h is a finite matrix-valued Radon measure concentrated on the $(p-1)$ -skeleton of $\mathcal{T}_h$ with atoms $A_F = [\nabla u_h]_F \otimes n_F |F|$; there are no atoms on lower-dimensional strata.
- (ii) **(Weak convergence.)** $\mu_h \Rightarrow \mu$ weakly as matrix-valued measures: for every $\varphi \in C_c^\infty(\Omega)$, $\langle \mu_h, \varphi \rangle \rightarrow \langle \mu, \varphi \rangle$. Moreover $\|\mu_h - \mu\|_{\text{KR}} \rightarrow 0$ in the Kantorovich–Rubinstein (flat) norm.
- (iii) **(Total variation: the honest statement.)** $\sup_h \|\mu_h\|_{\text{TV}} \leq C|u|_{C^2}|\Omega|$ (under quasi-uniformity) and $\liminf_h \|\mu_h\|_{\text{TV}} \geq \|\mu\|_{\text{TV}}$; but $\|\mu_h\|_{\text{TV}} \rightarrow \|\mu\|_{\text{TV}}$ is false in general — see Remark 4.2. It holds in the one-signed regime (convex or concave data restricted to one-dimensional parameter families), which is the regime of the parametric-LP value function and of κ^μ .
- (iv) **(No-mass-loss, correct form: L^1 reconstruction.)** On structured complexes, the per-cell reconstructed density $\rho_h(\text{cell}) = |\text{cell}|^{-1} \sum_{F \rightarrow \text{cell}} A_F$ converges to D^2u strongly in L^1 with rate $O(h)$ for $u \in C^3$; consequently $\|\rho_h\|_{L^1} \rightarrow \|\mu\|_{\text{TV}}$. On unstructured complexes the naive three-facet patch is not the dual cell; the general statement requires the discrete-duality (dual-diamond) pairing.
- (v) **(Resolution window.)** For the coarse-grained family $\mu_{h,\sigma} = \mu_h * \varphi_\sigma$: at fixed h , the limit $\sigma \rightarrow 0$ is the atomic measure; the smooth object exists only as a family member at matched resolution through the window $h \ll \sigma \ll L_{\text{var}}$ (the measured regime dial, §5.3).

Proof of (ii) (two lines). For $\varphi \in C_c^\infty(\Omega)$, $\langle D^2u_h, \varphi \rangle = \int_\Omega u_h D^2\varphi \text{dvol}$ by the definition of distributional derivatives (no boundary terms arise). Since $\|u - u_h\|_\infty \leq Ch^2|u|_{C^2}$ on shape-regular families, $\int u_h D^2\varphi \rightarrow \int u D^2\varphi = \langle \mu, \varphi \rangle$. The flat-norm statement follows by density of $\{c\varphi : \text{Lip}(\varphi) \leq 1\}$ in the KR dual cone and the uniform mass bound of (iii). $\square$

Remark 4.2 (Exact counterexample calculus). On the right-triangle mesh of $[0, 1]^2$ with Hessian $H = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, the per-cell atoms are $A_{\text{diag}} = bh^2 \mathbf{1}\mathbf{1}^\top$, $A_{\text{vert}} = (a-b)h^2 e_x \otimes e_x$, $A_{\text{horiz}} = (c-b)h^2 e_y \otimes e_y$; their *sum* is exactly $h^2 H$ (machine-verified to 0.0), while the entrywise total variation per cell is $|a-b| + |c-b| + 4|b|$ against the target $|a| + |c| + 2|b|$. Hence for $u = xy$ the ratio is exactly $3 - 1/n$, for the strictly convex $u = 2x^2 - xy + 2y^2$ it is $1.4 - 1/n$ (measured: 2.9922 and 1.3922 at $n = 128$), and for generic smooth maps 3.7–4.4. Total-variation convergence fails generically — including under convexity — while weak convergence holds at rate $O(h^2)$ throughout.

Proposition 4.3 (Two-layer separation and the sec law). *Let $K_h = \sum_v \text{defect}(v) \delta_v$ be the vertex angle-defect measure of the polyhedral graph $\{(x, u_h(x))\}$. Then K_h converges weakly to the Gauss-map area density*

$$\frac{\det D^2 u}{(1 + |\nabla u|^2)^{3/2}} \text{dvol} = K \sqrt{1 + |\nabla u|^2} \text{dvol}, \quad K = \frac{\det D^2 u}{(1 + |\nabla u|^2)^2},$$

i.e. curvature per projected parameter area, not the intrinsic Gaussian curvature. The bias factor is exactly $\sqrt{1 + |\nabla u|^2}$ (the sec law): machine-verified to 0.0% excess across 13 probes spanning tilts 0.11–2.83 and three tilt directions. Recovering intrinsic curvature requires the graph area correction.

Remark 4.4 (Consequences). (i) The codimension-one Hessian layer is the unbiased estimator and the canonical carrier for the bridge; (ii) Regge/angle-defect formulations need the area correction of Proposition 4.3; (iii) defects of a *fixed* complex (as measured in the M4b census) are scale-invariant $O(1)$ objects and are not contradicted by Proposition 4.3, which concerns refinement limits.

Proposition 4.5 (Regime dichotomy; M4a in closed form). *Let $B(\varepsilon)$ be the curvature mass captured by a loop of radius ε . For a C^2 map, $B(\varepsilon) = O(\varepsilon^2)$; for a piecewise-affine map whose kink complex meets the loop transversally, $B(\varepsilon) = O(\varepsilon)$. Measured slopes on the M4a design: 2.000 (smooth) and 1.00 (parametric FBA, 76-pair perturbation scan).*

4.1 [R-port] Bridge to parametric FBA

Conjecture 4.6 (mpLP refinement bridge). *Let $\{\Phi_n\}$ be a sequence of parametric-LP value functions (minima of tangent-plane families; θ in the right-hand side; nested outer approximations) converging uniformly to a C^2 limit f , with induced complexes $\mathcal{C}_n$. Then $\mu_n = D^2 \Phi_n$ converges weakly to $D^2 f$ dvol provided the complexes satisfy a per-cell discrete-duality identity (structured complexes) or a stable discrete-duality (dual-diamond) pairing. Along one-dimensional parameter cuts the conclusion holds unconditionally by monotone telescoping of the concave restriction.*

Remark 4.7 (Status). The one-dimensional cut case is machine-verified (independent-seed reproduction of the refinement experiment: folded Wasserstein distance $0.058 \rightarrow 0.0039$, mass ratio $0.939 \rightarrow 1.0009$ across $n = 4 \rightarrow 128$). The general complex case is open; the discrete-duality pairing stability is the identified missing assumption.

Remark 4.8 (Existence vs. resolution). Proposition 3.5 supplies existence of the curvature measure without refinement sequences, smoothing, or interpolation; Theorem B' governs what a discretely sampled trajectory can *resolve* of that measure, and at what rate. The two layers are complementary: the Alexandrov layer removes the extrinsic hypotheses, the refinement bridge keeps the quantitative rate.

Conjecture 4.9 (Value–flux layer relation). *Along generic one-dimensional cuts, the event structure of the flux map $v(\theta)$ is controlled by the facet structure of Φ : the count and mass of $D^2 v$ events between two parameter values are bounded by the crossed facets of $D^2(-\Phi)$ up to the degeneracy ratio observed at the iML1515 cut (12 flux events vs 1 value atom; value/flux mass ratio 1.7×10^{-6}), with the strain-circulation dichotomy (measured slopes 1.000 piecewise-affine vs 2.000 smooth) separating the regimes. If proved, the association of κ^μ with transcriptional response would promote from a flux-layer property to a value-function property.*

5 Computational validation

5.1 [E-M1] Active-set curvature verification

Twelve parameter sweeps (glucose, oxygen, eight gene knockdowns) on *E. coli* iML1515 with an iJO1366 replication, solved by a three-stage lexicographic pFBA engine (deterministic optimum;

fixed seeded tie-break weights after the discovery of pFBA vertex degeneracy). Result: 100.0000% of second-order response mass concentrates on operational active-set switches in every sweep that crosses critical-region boundaries (mass 0.934–1.0; AUC 0.83–1.00; MWU $p \leq 10^{-4}$); paths are piecewise affine to machine precision between events (relative segment residuals $\leq 8 \times 10^{-14}$); negative controls behave as theory dictates (unused-pathway knockdown: zero events, zero curvature; single-critical-region sweep: noise-level $D^2 \sim 10^{-11}$).

5.2 [E-M3] Double-knockout epistasis and path dependence

1,516 single knockouts and 2,779 double-knockout pairs across five panels. Epistasis $\varepsilon_{ij} = \kappa_{ij} - \kappa_i - \kappa_j$: all 40 synthetic-lethal pairs are isozyme redundancies (tktA/tktB, acnA/acnB, metE/metH, ...) with pure-emergence epistasis; $|\varepsilon_{ij}|$ aligns with active-set footprint overlap (Spearman $\rho_S = 0.865$, $J_{\text{support}} = 0.800$); sequential (L_1 -MOMA) knockouts open nonzero commutators in 25% of active pairs; closed genotype loops fail to return to the initial state in 66% of pairs (phenotypic memory; median holonomy ~ 110).

5.3 [E-M4] Regime dial

The smoothing identity $D^2(v * \phi_\sigma) = (D^2 v) * \phi_\sigma$ holds to kernel self-test error 1.2×10^{-6} ; measured crossover $\varepsilon^*/\sigma \in [2.45, 4.11]$ (median 3.1) separates the $O(\varepsilon^2)$ smooth regime from the $O(\varepsilon)$ discrete regime. The *M4a* scaling slope is 1.00 on the interacting stratum; the *M4b* census resolves the thin-wedge sliver structure (net measure jump 8.9 vs. nominal ± 1884.6).

5.4 [E-V1] The value function as canonical carrier

$\Phi = c_{\text{bio}}^\top v^*$ is single-valued with no tie-breaking; it is piecewise affine at 4.2×10^{-13} on the flux-event partition, carries one real atom $\Delta\Phi' = -0.006439$ at $t = 0.0358286$, and decouples from the flux-jump hierarchy (value/flux mass ratio 1.7×10^{-6}). Danskin: finite-difference shadow prices match envelope marginals to 6–7 digits. A controlled follower-variable LP reproduces the same structure: the value function stays affine and identical under two lexicographic tie-breaks while the follower’s second-difference mass flips $0.0100 \leftrightarrow 10^{-13}$ — the value layer is canonical (Proposition 3.5), the flux-strain layer is not (Conjecture 4.9).

6 Empirical association and protein-layer buffering

6.1 [E-V5] Transcriptional response under carbon depletion

On the deterministic lexicographic trajectory (E22 physiology, iJO1366), κ^μ (Definition 3.3) versus log-fold carbon-depletion response (M3D, $n = 424$ genes): Pearson $r = +0.395$ ($p = 2.6 \times 10^{-17}$), Spearman $\rho_S = +0.414$, partial $r = +0.269$ controlling reference level ($p = 1.8 \times 10^{-8}$), top-decile versus bottom-decile response 1.92 vs 0.89 (MWU 1.3×10^{-7}). Baseline reproduced to the digit ($r = +0.3739$, $n = 433$); metric invariance $\rho(\kappa^\mu, \kappa_V^{\text{lex}}) = 0.99998$ — both metrics sample the same flux-event structure along the same lexicographic trajectory, so the association is an event-structure property of the flux layer, not a metric artifact and not a value-function shadow.

6.2 [E-E27] Protein-layer null

Condition-dependent quantitative proteomics (22 conditions, triplicates, $n = 366$): protein-level correlation $r = -0.083$ against transcript $+0.419$ on the same genes. The transcript-level association does not propagate to the protein layer; buffering at translation is the consistent model, positioning Kochanowski et al. as prior art for the protein-layer null.

7 Discussion: one measure, several resolutions

The three sensitivity objects of the earlier literature — the geometric κ_V , the flux-statistical κ_{flux} , and the time-course κ_{time} — are one measure at different resolutions: the atomic measure of Definition 3.1, its coarse-grainings $\mu * \phi_\sigma$, and its integrals along parameter trajectories (κ^μ). The smooth member exists only through the window $h \ll \sigma \ll L_{\text{var}}$; the $\sigma \rightarrow 0$ limit is the atomic object. Open items, demoted to their honest status: Conjecture 4.6 (general complex stability), the formal identity $\kappa_{\text{flux}} = \mathcal{F}[\mu]$ (decoupled from the empirical association by the metric-invariance result), E28 under the (ε, σ) design law, and E32 (event-measure stabilization — the L^1 -reconstruction regime of Theorem 4.1(iv)).

Limitations. Correlational design; single-organism scope; operational (not combinatorial) active sets; lexicographic tie-breaking is an engine convention (the value-function carrier is immune); categorical and homotopy-theoretic material of the predecessor manuscript is retained only where it carries falsifiable content, with the remainder moved to a companion document.

8 Methods

Engine. Three-stage lexicographic pFBA on HiGHS (lexicographic order: biomass, then ℓ_1 flux, then seeded deterministic weights); vertex degeneracy of plain pFBA documented (0.69 flux flips between warm-started solves) and resolved deterministically.

Data. M3D microarray panels (carbon-depletion contrast); PRECISE 2.0 evaluated (no carbon-depletion class, documented); Schmidt et al. 2016 condition-dependent proteome. SHA-256 provenance manifests deposited.

Reproducibility. All experiments deposit artifacts (npz/csv/json) with scripts; independent-seed reproductions recorded for the refinement prototype and the stress batteries.

A Layer-1 porting map

v2 section	v21 source (frozen; do not modify)
§1	v21 sec 1 (rewritten to the single story)
§2	v21 sec 2 (subset: active-set notation only)
§3	v21 active-set stratification defs; new D1/D2
§4	new (Theorem B'; replaces the smooth bridge)
§5	M1/M3/M4 report bodies
§6	v21 E22–E27 sections
§7	v21 discussion, rewritten as resolution statement
§8	v21 methods; engine documentation
companion	categorical/HoTT sections of v21 (trimmed subset)